

## 1.4 Calculus

The constant of integration has been omitted.

| $\int y \, dx$                                                                                        | $y$                       | $\frac{dy}{dx}$                    |
|-------------------------------------------------------------------------------------------------------|---------------------------|------------------------------------|
| $\left. \begin{array}{ll} \frac{x^{n+1}}{n+1} & n \neq -1 \\ \log_e(x) & n = -1 \end{array} \right\}$ | $x^n$                     | $nx^{n-1}$                         |
| $x \log_e(x) - x$                                                                                     | $\log_e(x)$               | $\frac{1}{x}$                      |
| $\frac{1}{a} e^{ax}$                                                                                  | $e^{ax}$                  | $ae^{ax}$                          |
| $\frac{a^x}{\log_e(a)}$                                                                               | $a^x \quad (a > 0)$       | $a^x \log_e(a)$                    |
| $-\frac{1}{a} \cos(ax)$                                                                               | $\sin(ax)$                | $a \cos(ax)$                       |
| $\frac{1}{a} \sin(ax)$                                                                                | $\cos(ax)$                | $-a \sin(ax)$                      |
| $\frac{1}{a} \log_e[\sec(ax)]$                                                                        | $\tan(ax)$                | $a \sec^2(ax)$                     |
| $\log_e \left[ \tan \left( \frac{x}{2} \right) \right]$                                               | $\operatorname{cosec}(x)$ | $-\operatorname{cosec}(x) \cot(x)$ |
| $\log_e[\sec(x) + \tan(x)]$                                                                           | $\sec(x)$                 | $\sec(x) \tan(x)$                  |
| $\log_e[\sin(x)]$                                                                                     | $\cot(x)$                 | $-\operatorname{cosec}^2(x)$       |
| $\cosh(x)$                                                                                            | $\sinh(x)$                | $\cosh(x)$                         |
| $\sinh(x)$                                                                                            | $\cosh(x)$                | $\sinh(x)$                         |
| $\sin^{-1}(x)$                                                                                        | $\frac{1}{\sqrt{1-x^2}}$  |                                    |
| $\cos^{-1}(x)$                                                                                        | $\frac{1}{\sqrt{x^2-1}}$  |                                    |
| $\tan^{-1}(x)$                                                                                        | $\frac{1}{1+x^2}$         |                                    |
| $\sec^{-1}(x)$                                                                                        | $\frac{1}{x\sqrt{x^2-1}}$ |                                    |
| $\left. \begin{array}{l} \sinh^{-1}(x) \\ \log_e(x + \sqrt{x^2+1}) \end{array} \right\}$              | $\frac{1}{\sqrt{x^2+1}}$  |                                    |
| $\left. \begin{array}{l} \cosh^{-1}(x) \\ \log_e(x + \sqrt{x^2-1}) \end{array} \right\}$              | $\frac{1}{\sqrt{x^2-1}}$  |                                    |
| $\tanh^{-1}(x)$                                                                                       | $\frac{1}{1-x^2}$         |                                    |

| $\int y \, dx$                                                                                    | $y$                            |
|---------------------------------------------------------------------------------------------------|--------------------------------|
| $\sin^{-1} \left( \frac{x}{a} \right)$                                                            | $\frac{1}{\sqrt{a^2 - x^2}}$   |
| $\cos^{-1} \left( \frac{x}{a} \right)$                                                            | $-\frac{1}{\sqrt{a^2 - x^2}}$  |
| $\tan^{-1} \left( \frac{x}{a} \right)$                                                            | $\frac{a}{a^2 + x^2}$          |
| $\sec^{-1} \left( \frac{x}{a} \right)$                                                            | $-\frac{a}{x\sqrt{a^2 - x^2}}$ |
| $\sinh^{-1} \left( \frac{x}{a} \right)$ or $\log_e \left( \frac{x + \sqrt{x^2 + a^2}}{a} \right)$ | $\frac{1}{\sqrt{a^2 + x^2}}$   |
| $\cosh^{-1} \left( \frac{x}{a} \right)$ or $\log_e \left( \frac{x + \sqrt{x^2 - a^2}}{a} \right)$ | $-\frac{1}{\sqrt{a^2 - x^2}}$  |
| $\tanh^{-1} \left( \frac{x}{a} \right)$                                                           | $\frac{a}{a^2 - x^2}$          |

### Multiplication by a scalar

If  $y = ku(x)$  where  $k$  is a scalar then

$$\frac{dy}{dx} = k \frac{du}{dx}$$

### Product rule

If  $y = u(x)v(x)$  then

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

### Quotient rule

If  $y = \frac{u(x)}{v(x)}$  then

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

### Chain rule

If  $y$  is a function of  $u$  and  $u$  is the function of  $x$  then

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

### Sum/difference rules

$$\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$$